

§ Lecture 12.0

Thursday, 24 September 2026

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Lemma (Non zero reals): let $x \neq 0$. Then there exists a Cauchy sequence

$(a_n)_{n \geq 1}$ of rationals that is bounded away from zero such that

$$x = \lim_{n \rightarrow \infty} a_n.$$

Proof: Let $x = \lim_{n \rightarrow \infty} b_n \neq \lim_{n \rightarrow \infty} 0$

i.e. $(b_n)_{n \geq 1}$ is not equivalent $(0)_{n \geq 1}$.

$$\Rightarrow \exists \varepsilon_0 > 0 \quad \forall N \geq 1 \quad \exists n \geq N \text{ s.t. } |b_n| > \varepsilon_0.$$

Fix such ε_0 . Then for all $N \geq 1 \exists n \geq N$ s.t. $|b_n| > \varepsilon_0 \rightarrow \textcircled{1}$

Since (b_n) is Cauchy

$$\forall \varepsilon > 0 \exists N_0 \geq 1 \text{ s.t. } |b_m - b_n| \leq \varepsilon \rightarrow \textcircled{2}$$

Apply (1) for N_0 , $\exists n_0 \geq N_0$ s.t. $|b_{n_0}| > \varepsilon_0$.

In $\textcircled{2}$ choose $\varepsilon = \varepsilon_0/2$, then

$$|b_m - b_{n_0}| \leq \varepsilon_0/2 \quad \& \quad |b_{n_0}| > \varepsilon_0 \quad \forall m \geq N_0.$$

$$\Rightarrow b_{n_0} - \varepsilon_0/2 \leq b_m \leq b_{n_0} + \varepsilon_0/2 \quad \forall m \geq N_0$$

$$\Rightarrow \varepsilon_0/2 \leq b_m \leq -\varepsilon_0/2 \quad \forall m \geq N_0$$

$$\Rightarrow |b_m| \geq \varepsilon_0/2 \quad \forall m \geq N_0.$$

Choose $(a_n) = (\varepsilon_0/2, \dots, \varepsilon_0/2, b_{N_0}, \dots)$

$$a_n = \begin{cases} \varepsilon_0/2 & \forall n < N_0 \\ b_n & \forall n \geq N_0 \end{cases}$$

(a_n) & (b_n) are equivalent.

$$\Rightarrow x = \lim_{n \rightarrow \infty} a_n, \text{ where } |a_n| \geq \varepsilon_0/2 \quad \forall n \geq 1.$$

Lemma: Suppose $(a_n)_{n \geq 1}$ is a Cauchy sequence that is bounded away from zero. Then $(a_n^{-1})_{n \geq 1}$ is a Cauchy sequence.

Proof: Since (a_n) is a Cauchy sequence that is bounded away from zero, we have

$$(i) \exists c > 0 \text{ s.t. } |a_n| \geq c \quad \forall n \geq 1. \rightarrow \textcircled{1}$$

$$(ii) \forall \delta > 0 \exists N \geq 1 \text{ s.t. } \forall m, n \geq N : |a_m - a_n| \leq \delta \rightarrow \textcircled{2}$$

Fix $\varepsilon > 0$. Consider

$$\begin{aligned} |a_m^{-1} - a_n^{-1}| &= \left| \frac{a_n - a_m}{a_n a_m} \right| \\ &\leq c^2 |a_n - a_m| \end{aligned} \quad \text{Using } \textcircled{1}$$

Applying $\textcircled{2}$ for $\delta = \varepsilon/c^2$

$$\Rightarrow |a_m^{-1} - a_n^{-1}| \leq \varepsilon \quad \forall m, n \geq N.$$

Since $\varepsilon > 0$ is arbitrary $\Rightarrow (a_n^{-1})$ is a Cauchy sequence.

Def (Reciprocal of reals): let $x \neq 0$ and $x = \lim_{n \rightarrow \infty} a_n$, where (a_n) is a "bounded away from zero" sequence. Then

$$x^{-1} = \lim_{n \rightarrow \infty} a_n^{-1}.$$

Well definedness: $\textcircled{1}$ Indeed (a_n^{-1}) is Cauchy.

$$\textcircled{2} \text{ If } \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n \text{ then } \lim_{n \rightarrow \infty} a_n^{-1} = \lim_{n \rightarrow \infty} b_n^{-1}.$$

Proof: Trick:

$$\begin{aligned} P &= \lim_{n \rightarrow \infty} a_n^{-1} \lim_{n \rightarrow \infty} a_n \lim_{n \rightarrow \infty} b_n^{-1} \\ &= \lim_{n \rightarrow \infty} a_n^{-1} a_n b_n^{-1} = \lim_{n \rightarrow \infty} b_n^{-1} \\ P &= \lim_{n \rightarrow \infty} a_n^{-1} b_n b_n^{-1} = \lim_{n \rightarrow \infty} a_n^{-1}. \end{aligned}$$

$$* \quad x, y \in \mathbb{R}, \text{ with } y \neq 0 \text{ then } x/y = x \cdot y^{-1}.$$

How to define positive and negative real numbers?

Consider sequences: $(10^{-n})_{n \geq 1}$ $(-10^{-n})_{n \geq 1}$ $(-10^{-n})_{n \geq 1}$

Has all positive Has mixed All negative

But all of them define real numbers 0.

Def (Positively and negatively bounded from away sequences):

Let $(a_n)_{n \geq 1}$ be a Cauchy sequence of rationals. Then

$\textcircled{1}$ If $\exists c > 0$ s.t. $a_n \geq c \quad \forall n \geq 1$: Positively bounded away from zero

$\textcircled{2}$ If $\exists -c < 0$ s.t. $a_n \leq -c \quad \forall n \geq 1$: Negatively " " " "

Positive/Negative reals: $x \in \mathbb{R}$ is said to be positive/negative iff

there is a Cauchy sequence $(a_n)_{n \geq 1}$ which is positively/negatively bounded away from zero s.t. $x = \lim_{n \rightarrow \infty} a_n$.

§ Lecture 12.1

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Ordering of reals.

Claim 1: For every real x , exactly one of the statements is true. (i) $x=0$ (ii) x is positive (iii) x is negative.

Proof: At least one of them is true.

If $x=0$, there is nothing to prove. So assume $x \neq 0$.

Then there exist a sequence $(a_n)_{n \geq 1}$ that is bounded away from

zero s.t. $x = \lim_{n \rightarrow \infty} a_n$.

so $|a_n| \geq c \quad \forall n \geq 1$ for some $c > 0$.

$$\Leftrightarrow a_n \geq c \text{ or } a_n \leq -c \quad \forall n \geq 1.$$

Can they be mixed?

Note (a_n) is a Cauchy sequence.

$$\forall \delta > 0 \quad \exists N \geq 1 \text{ s.t. } |a_m - a_n| \leq \delta \quad \forall m, n \geq N. \rightarrow (1)$$

Claim: Either $a_n \geq c \quad \forall n \geq N$ or $a_n \leq -c \quad \forall n \geq N$.

Proof. Let $i, j \geq N$, and $a_i \geq c$ & $a_j \leq -c$ (wlog).

$$\text{then } a_i - a_j \geq 2c \quad \text{This contradicts eqn (1).}$$

Now let $a_n \geq c \quad \forall n \geq N$.

$$\text{Define } b_n = \begin{cases} c & \forall n < N \\ a_n & \forall n \geq N \end{cases}$$

(a_n) and (b_n) are equivalent. and (b_n) is positively bounded away.

$\Rightarrow x$ is positive.

For $a_n \leq -c \quad \forall n \geq N$, we can find an equivalent negatively bounded

away sequence $\Rightarrow x$ is negative.

At most one of the statements is true.

$\rightarrow$ Positive and zero.

$$x = \lim_{n \rightarrow \infty} a_n \text{ with } a_n \geq c > 0 \quad \& \quad x = 0 = \lim_{n \rightarrow \infty} 0.$$

$\Rightarrow (a_n)$ is equivalent to (0) .

$$\Rightarrow \forall \varepsilon > 0 \quad \exists N \geq 1 \text{ s.t. } |a_n| \leq \varepsilon.$$

$$\text{Take } \varepsilon = c/2 \Rightarrow |a_n| \leq c/2 < c \\ \Rightarrow c \leq c. \text{ Impossible.}$$

$\rightarrow$ Positive and negative.

$$x = \lim_{n \rightarrow \infty} a_n \quad a_n \geq c_1 > 0 \quad \forall n$$

$$x = \lim_{n \rightarrow \infty} b_n \quad b_n \leq -c_2 < 0 \quad \forall n.$$

$$\text{Then } a_n - b_n \geq c_1 + c_2 \quad \forall n.$$

but $(a_n), (b_n)$ are equivalent $\Rightarrow \forall \delta > 0 \quad \exists N$ s.t. $|a_n - b_n| \leq \delta$.

$$\text{Pick } \delta = (c_1 + c_2)/2$$

$$\Rightarrow \frac{c_1 + c_2}{2} \leq |a_n - b_n| \geq c_1 + c_2 > \frac{c_1 + c_2}{2}$$

$$\Rightarrow \frac{c_1 + c_2}{2} > \frac{c_1 + c_2}{2} \quad (\text{Impossible})$$

* Negative and zero. Similar to positive and zero.

$$\text{Def. (1)} \quad |x| = \begin{cases} x & \text{if } x \text{ positive} \\ 0 & \text{if } x = 0 \\ -x & \text{if } x \text{ negative.} \end{cases}$$

$$(2) \quad x > y \Leftrightarrow x - y \text{ is positive.}$$

$$x \geq y \Leftrightarrow x > y \text{ or } x = y.$$

Numerous other claims:

$$(1) \quad \text{if } x > 0 \text{ then } x^{-1} > 0.$$

$$(2) \quad \text{if } x > 0, y > 0 \text{ then } x > y \Leftrightarrow x^{-1} < y^{-1}.$$

Claim: Let $(a_n)_{n \geq 1}$ be a Cauchy sequence of nonnegative rationals then

$$x = \lim_{n \rightarrow \infty} a_n \geq 0.$$

Proof: By contradiction.

Suppose $x < 0$. Then there is a Cauchy sequence $(b_n)_{n \geq 1}$ that is

negatively bounded away from zero, i.e. $b_n \leq -c < 0$ such that

$\rightarrow (1)$

$$x = \lim_{n \rightarrow \infty} b_n$$

$$\text{On the other hand, } x = \lim_{n \rightarrow \infty} a_n \text{ with } a_n \geq 0 \quad \forall n \geq 1. \rightarrow (2)$$

The equivalence of (a_n) and $(b_n) \Rightarrow \forall \delta > 0 \quad \exists N \geq 1$ s.t. $|a_n - b_n| \leq \delta \quad \forall n \geq N. \rightarrow (3)$

Consider $|a_n - b_n|$. From (3)

$$|a_n - b_n| \leq \delta/2 < c \text{ But } a_n - b_n \geq c > 0$$

$$\Rightarrow |a_n - b_n| \geq c.$$

$$\Rightarrow c \leq |a_n - b_n| < c \Rightarrow c < c \text{ contradiction.}$$

Claim: If (a_n) and (b_n) are two Cauchy sequences with

$$a_n \geq b_n. \text{ Then } \lim_{n \rightarrow \infty} a_n \geq \lim_{n \rightarrow \infty} b_n.$$

Proof: Define $y_n = a_n - b_n \geq 0$, from previous claim $\lim_{n \rightarrow \infty} y_n \geq 0$

$$\Rightarrow \lim_{n \rightarrow \infty} a_n \geq \lim_{n \rightarrow \infty} b_n.$$

Caution: $a_n > b_n \not\Rightarrow \lim_{n \rightarrow \infty} (a_n - b_n) > 0. !!$

§ Lecture 12.2

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Lemma (Bounding of reals by rationals).

Let $x > 0$. Then there exists a positive rational q s.t. $q \leq x$,

and there exists a positive integer N s.t. $x \leq N$.

Proof: Since $x > 0$, we can write

$x = \lim_{n \rightarrow \infty} a_n$, where $(a_n)_{n \geq 1}$ is a Cauchy sequence with

$$a_n \geq q > 0 \quad \forall n \geq 1.$$

Since (a_n) is Cauchy $\exists$ rational $\gamma > 0$ s.t. $|a_n| \leq \gamma \quad \forall n \geq 1$.
 $\Rightarrow a_n \leq \gamma \quad \forall n \geq 1$

That is $q \leq a_n \leq \gamma \leq N$
 $\uparrow$ N is positive as q is positive.

$$\Rightarrow q \leq a_n \leq N$$

Taking formal limits $\lim_{n \rightarrow \infty} q \leq \lim_{n \rightarrow \infty} a_n \leq \lim_{n \rightarrow \infty} N$

$$\Leftrightarrow q \leq x \leq N.$$

Archimedean property of reals. Let $x, \varepsilon > 0$ be real numbers. Then

there exists a positive integer M s.t. $M\varepsilon > x$.

Proof: $\frac{x}{\varepsilon}$ is a positive real number. $\Rightarrow \exists$ integer $N > 0$ s.t.

$$\frac{x}{\varepsilon} \leq N < N+1 \Rightarrow (N+1)\varepsilon > x. \text{ Call } M = N+1.$$

Remark: No matter how small is ε and how large is x , if

one keeps adding ε , it will exceed x .